FAULT CURRENT STUDIO - ENGINEERING REPORT
BROWSER EXPORT - ENGINEERING REVIEW REQUIRED

===== SUMMARY =====
Project: Fault Current Studio advanced study
Method: phasorduty
Model status: DOCUMENTED_SYMMETRICAL_PHASOR_THERMAL_DUTY
Captured UTC: 2026-10-26T06:04:43.394Z
Warning: ENGINEERING SCREENING ONLY ? independently verify actual manufacturer, utility, installation, e
quipment settings and standards edition before professional use.
Report data provenance: Entered study inputs, isolated calculated method outputs. No external OEM certif
ication or auto-selected breaker/PPE.

===== STUDY INPUTS =====
Input path: durationS | Entered value: 0.1
Input path: sources[0].id | Entered value: UTILITY
Input path: sources[0].sourceReference | Entered value: ILLUSTRATIVE FAULT PHASOR TRACE; replace with OE
M/network time study
Input path: sources[0].samples[0].timeS | Entered value: 0
Input path: sources[0].samples[0].symmetricalRmsA | Entered value: 14000
Input path: sources[0].samples[0].phaseDeg | Entered value: 0
Input path: sources[0].samples[1].timeS | Entered value: 0.05
Input path: sources[0].samples[1].symmetricalRmsA | Entered value: 10000
Input path: sources[0].samples[1].phaseDeg | Entered value: 0
Input path: sources[0].samples[2].timeS | Entered value: 0.1
Input path: sources[0].samples[2].symmetricalRmsA | Entered value: 6000
Input path: sources[0].samples[2].phaseDeg | Entered value: 0
Input path: sources[1].id | Entered value: GENERATOR
Input path: sources[1].sourceReference | Entered value: ILLUSTRATIVE GENERATOR TRACE; replace with OEM d
ecrement and case
Input path: sources[1].samples[0].timeS | Entered value: 0
Input path: sources[1].samples[0].symmetricalRmsA | Entered value: 4000
Input path: sources[1].samples[0].phaseDeg | Entered value: 30
Input path: sources[1].samples[1].timeS | Entered value: 0.1
Input path: sources[1].samples[1].symmetricalRmsA | Entered value: 0
Input path: sources[1].samples[1].phaseDeg | Entered value: 30

===== EQUIPMENT DUTY =====
Calculated output path: status | Value: DOCUMENTED_SYMMETRICAL_PHASOR_THERMAL_DUTY
Calculated output path: durationS | Value: 0.1
Calculated output path: sourceCount | Value: 2
Calculated output path: sourceReferences[0].id | Value: UTILITY
Calculated output path: sourceReferences[0].reference | Value: ILLUSTRATIVE FAULT PHASOR TRACE; replace
with OEM/network time study
Calculated output path: sourceReferences[1].id | Value: GENERATOR
Calculated output path: sourceReferences[1].reference | Value: ILLUSTRATIVE GENERATOR TRACE; replace wit
h OEM decrement and case
Calculated output path: totalI2tA2S | Value: 14992648.497156125
Calculated output path: equivalentSymmetricalRmsA | Value: 12244.447107630513
Calculated output path: maximumInstantaneousSymmetricalRmsA | Value: 17578.249208150883
Calculated output path: timeAtMaximumS | Value: 0
Calculated output path: individualSourceI2tA2S.UTILITY | Value: 10533333.333333334
Calculated output path: individualSourceI2tA2S.GENERATOR | Value: 533333.3333333334
Calculated output path: intervalCount | Value: 2
Calculated output path: approved | Value: false
Calculated output path: method | Value: Exact |sum complex RMS phasors|2 integral under explicitly assum
ed linear complex-phasor interpolation between provided time-stamped points
Calculated output path: limitations[0] | Value: Fundamental-frequency symmetrical RMS phasor only: no as

ymmetrical DC offset, high harmonics, CT/relay response or waveform integration
Calculated output path: limitations[1] | Value: Phasor time traces are externally supplied and must be a
ppropriate to the same fault case and network
Calculated output path: limitations[2] | Value: Time values in traces must include all switching/discont
inuities or separate sequences; interpolation is NOT a generator/IBR physics model
Calculated output path: limitations[3] | Value: Not equivalent IEC 60909/ANSI short-time duty, peak maki
ng or breaking-capacity certification

===== ASSUMPTIONS & QA =====
Evidence / limitation: Release status | Value: BOUNDED METHOD, NOT A STANDARDS/DEVICE APPROVAL
Evidence / limitation: Warning | Value: ENGINEERING SCREENING ONLY ? independently verify actual manufac
turer, utility, installation, equipment settings and standards edition before professional use.
Evidence / limitation: Study method | Value: phasorduty
Evidence / limitation: Timestamp | Value: 2026-10-26T06:04:43.394Z
Evidence / limitation: outputs.limitations[0] | Value: Fundamental-frequency symmetrical RMS phasor only
: no asymmetrical DC offset, high harmonics, CT/relay response or waveform integration
Evidence / limitation: outputs.limitations[1] | Value: Phasor time traces are externally supplied and mu
st be appropriate to the same fault case and network
Evidence / limitation: outputs.limitations[2] | Value: Time values in traces must include all switching/
discontinuities or separate sequences; interpolation is NOT a generator/IBR physics model
Evidence / limitation: outputs.limitations[3] | Value: Not equivalent IEC 60909/ANSI short-time duty, pe
ak making or breaking-capacity certification